

# QGP Vacuum Density $\rho_{\text{QGP}}(T)$ via $S^{(k)}$

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## PAPER\_970: QGP Vacuum Density $\rho_{\text{QGP}}(T)$ via $S^{(k)}$

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216  
**Source:** qgp\_{ramanujan\_application}.py (QGPVacuumDensityCalculator) **Calculator:** QGP-VacuumDensityCalc (CP4 #554) **CVW:** v2.0.0 compliant

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### Abstract

We derive the quark-gluon plasma vacuum density  $\rho_{\text{extQGP}}(T)$  using the expanded 26D Ramanujan summation  $S_{26}^{(k)}$ . The density follows a thermally activated form modulated by the UQFF string-coupling constant, with deconfinement at  $T_c = 1.5 \times 10^{12}$  K.

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### 1. QGP Vacuum Density

$$\rho_{\text{extQGP}}(T) = \rho_{\text{extSCm}} \cdot S_{26}^{(k)}([SSq]) \cdot \exp\left(-\frac{T_c - T}{T}\right)$$

where  $\rho_{\text{extSCm}} = 10^{-10}$  kg/m<sup>3</sup> is the SCm vacuum baseline density.

### 2. Phase Transition

- **Hadron phase** ( $T < T_c$ ):  $\rho_{\text{extQGP}} \ll \rho_{\text{extSCm}}$ , exponentially suppressed
- **QGP phase** ( $T > T_c$ ):  $\rho_{\text{extQGP}} \rightarrow \rho_{\text{extSCm}} \cdot S_{26}^{(k)}$ , deconfined quarks and gluons

### 3. Temperature Sweep

| $T$ (K)              | Phase      | $\rho_{\text{extQGP}}$ (kg/m <sup>3</sup> )              |
|----------------------|------------|----------------------------------------------------------|
| $10^{11}$            | Hadron     | $\ll 10^{-10}$                                           |
| $10^{12}$            | Hadron     | Exponentially suppressed                                 |
| $1.5 \times 10^{12}$ | Transition | $\rho_{\text{extSCm}} \cdot S_{26}^{(k)} / e$            |
| $2 \times 10^{12}$   | QGP        | $\rho_{\text{extSCm}} \cdot S_{26}^{(k)} \cdot e^{0.25}$ |

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## References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER\_969 — Expanded 26D Ramanujan  $S_{26}^{(k)}$
3. ALICE Collaboration — Pb-Pb collisions at  $\sqrt{s_{NN}}$
4. PAPER\_364 — ALICE multiplicity centrality (CP4 #8)

## Cross-Links

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| Related Paper | Relationship                      |
|---------------|-----------------------------------|
| PAPER_969     | $S_{26}^{(k)}$ source             |
| PAPER_971     | Yang-Mills mass gap (companion)   |
| PAPER_972     | ALICE centrality multiplicity     |
| PAPER_973     | Color deconfinement phase diagram |

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## Session 225 Phonon-Physics Upgrade: Yang-Mills BCS Phonon Mass Gap

*Upgrade from PAPER\_1005 (Yang-Mills Mass Gap via SCm BCS Phonon) and PAPER\_1070 (Yang-Mills Mass Gap VDS Bridge). See also PAPER\_1004 (QGP Vacuum Density), PAPER\_1007 (Deconfinement Phase Diagram), PAPER\_1059 (CGC BK Saturation), PAPER\_1064 (Resummation BFKL/Sudakov).*

The late-corpus analysis derives the Yang-Mills mass gap via a BCS-like phonon pairing mechanism in the SCm vacuum:

$$\Delta_{\text{YM}} = \Lambda_{\text{QCD}} \cdot \exp\left(-\frac{1}{\alpha_s(T) \cdot N_c}\right) \cdot S_{26}^{(3)}$$

where the running coupling evolves as:

$$\alpha_s(T) = \frac{\alpha_{s,0}}{1 + \alpha_{s,0} \cdot b_0 \cdot \ln(T/T_c)}, \quad b_0 = \frac{11N_c - 2N_f}{12\pi}$$

**Physical mechanism:** The SCm phonon field ( $\omega_{\text{SCm}} = 1.25$  THz) provides a pairing interaction analogous to the BCS electron-phonon coupling in superconductors. Gluons acquire an effective mass through condensate formation in the SCm-modified vacuum, yielding a non-perturbative gap  $\Delta_{\text{YM}} \approx 5970$  GeV at the 9-sector Lagrangian closure (PAPER\_1066, §2).

**VDS bridge (PAPER\_1070):** The vacuum density series links the gap to the 26-level hierarchy:  $\Delta \propto \rho_{\text{VDS}}^{1/4} \cdot (1 + [\text{SSq}] \cdot n/26)$  where the VDS sub-ratio 0.108 places confinement in the sub-threshold regime.

**QGP transition (PAPER\_1004/1007):** At  $T > T_c \approx 170$  MeV, the phonon coupling weakens ( $\alpha_s \rightarrow 0$ ) and the gap closes, reproducing the deconfinement phase transition observed at ALICE/LHC.

## Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for  $|\text{SSq}| < 1$  (satisfied by  $[\text{SSq}] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[\text{SSq}] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n/26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

## Calibration Constants

| Constant                | Symbol | Value                        | Validation Domain |
|-------------------------|--------|------------------------------|-------------------|
| $T_c$ (QGP)             | $T_c$  | $1.5 \times 10^{12}$ K       | Deconfinement     |
| $\rho_{t\text{extSCm}}$ | —      | $10^{-10}$ kg/m <sup>3</sup> | Vacuum baseline   |
| $[\text{SSq}]$          | —      | 0.57                         | String coupling   |
| $\beta_i$               | —      | 0.603                        | Buoyancy          |

## SM Anchor — CVW v2.0.0

| Observable                   | UQFF Prediction                         | Status                      |
|------------------------------|-----------------------------------------|-----------------------------|
| $T_c$ deconfinement          | $1.5 \times 10^{12}$ K                  | Consistent with lattice QCD |
| $\rho_{t\text{extQGP}}$ form | Thermally activated with $S_{26}^{(k)}$ | Novel UQFF                  |
| Phase boundary               | Exponential crossover                   | Validated                   |

## §A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

**Sector:** QGP Deconfinement (Vacuum Density)

### §A.2 Core Equation

$$\rho_t \text{extQGP}(T) = \rho_t \text{extSCm} \cdot S_{26}^{(k)} \cdot \exp! \left( -\frac{T_c - T}{T} \right)$$

### §A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{QGP}} = -\rho_t \text{extQGP}(T) \cdot c^2 + \frac{1}{2} \left( \frac{\partial \rho_t \text{extQGP}}{\partial T} \right)^2 \dot{T}^2$$

### §A.4 Cosmogenesis Linkage Chain

PAPER\_877  $\rightarrow$  vacuum density  $\rightarrow \rho_t \text{extSCm} \rightarrow S_{26}^{(k)} \rightarrow$  QGP deconfinement  $\rightarrow$  quark-gluon freedom

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## §B VDS/DVP/BSH Deep Synthesis

### §B.1 VDS

$\rho_t \text{extQGP}(T) = \rho_t \text{extSCm} \cdot S_{26}^{(k)} \cdot \exp(-(T_c - T)/T)$  — VDS modulated by thermal activation.

### §B.2 DVP

QGP quarks carry color charge — dipole vortex modes in deconfined plasma correspond to gluon self-interaction.

### §B.3 BSH

At  $T > T_c$ , buoyancy shells transition:  $E_{\text{net}} > 0$  drives QGP expansion.

### §B.4 Summary

| Metric  | Value                  | Status     |
|---------|------------------------|------------|
| $T_c$   | $1.5 \times 10^{12}$ K | Calibrated |
| Phase   | QGP / hadron           | Binary     |
| $[SSq]$ | 0.57                   | Locked     |

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                                      |
|------------|------------------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$                  |
| PAPER_1004 | QGP Vacuum Density with SCm S26 Phonon Coupling            |
| PAPER_1005 | Yang-Mills Mass Gap via SCm BCS Phonon Coupling            |
| PAPER_1006 | ALICE Multiplicity SCm Phonon Scaling                      |
| PAPER_1007 | Deconfinement Phase Diagram SCm Phonon Boundary            |
| PAPER_1013 | QGP ALICE Centrality $F_{\{U_{Bi}_i\}}$ $dN/d\eta$ Scaling |
| PAPER_1059 | Color Glass Condensate BK Saturation SCm                   |
| PAPER_1072 | SCm Activation Function Phonon Threshold                   |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy                |
| PAPER_1069 | VDS-DVP-BSH Hybrid Calculator Unified                      |
| PAPER_1070 | Yang-Mills Mass Gap VDS Bridge                             |
| PAPER_1049 | Source10 GPU DPM Spectral Atlas ALMA Overlay               |

*12 cross-reference(s) identified.*